NutriTrack AI

Native Mobile App UI & Feature Reference Manual

Comprehensive Visual Specs, Screen Workflows, and Architecture for React Native & Expo

Target Platforms:	Android 11+ & iOS 15+ (React Native 0.76+ / Expo SDK 52+)
Source Framework:	Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS
Machine Learning:	OpenCLIP ViT-B-32 Zero-Shot Food Vision + FastAPI Backend
Primary Colors:	Emerald Green (#059669), Dark Slate (#0f172a), Slate (#1e293b)
Author / Project:	Google Antigravity Engineering Handover to Replit AI

EXECUTIVE SUMMARY FOR REPLIT AI:
This visual manual documents every existing user interface of NutriTrack AI captured directly from the live working application. For each screen, both desktop and mobile viewports are provided alongside detailed technical annotations covering: (1) UI component hierarchy, (2) user interactions and gesture behaviors, (3) backend API integration endpoints, (4) mathematical formulas and state keys, and (5) exact React Native component translation guidelines.

1. Design Tokens & Visual Architecture

NutriTrack utilizes an active health aesthetic balancing deep, calm dark slate tones with vibrant emerald highlights, conveying clinical trust and dynamic wellness.

Token Name	Hex Code	CSS / Tailwind Class	Mobile Usage
Primary Brand	#059669	emerald-600	Main CTAs, active tab icons, gauge highlights
Primary Glow	#10b981	emerald-500	Success badges, completion rings, toggles
Dark Base	#0f172a	slate-900	Screen root background, safe-area headers
Elevated Card	#1e293b	slate-800	Surface cards, modals, input containers
Border / Divider	#334155	slate-700	Card outlines, list separators, chip outlines
Alert Amber	#f59e0b	amber-500	Calorie surplus warnings, missing log reminders
Alert Rose	#f43f5e	rose-500	Severe over-budget flags, error states
Hydration Sky	#0284c7	sky-600	Water tracker bar, fluid intake badges

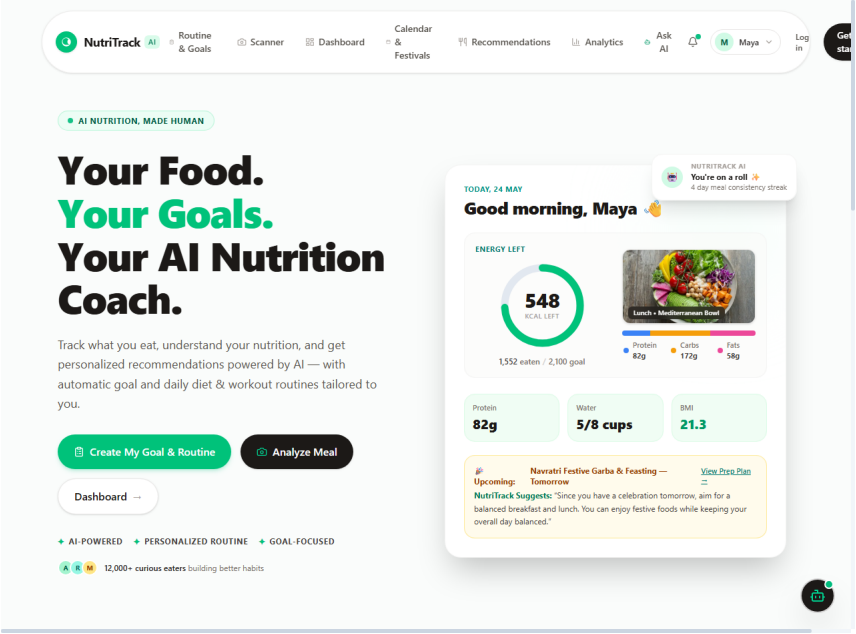
Typography & Elevation Standards

- **Font Family:** System Sans (Inter / SF Pro Display on iOS, Roboto on Android).
- **Headings:** Bold weight (700/800), tight letter-spacing (-0.02em).
- **Body Text:** Regular weight (400) on white/slate-100; Muted text on slate-400.
- **Card Elevation:** Subtle drop shadows on iOS (`shadowOpacity: 0.15, shadowRadius: 10`), `elevation: 4` on Android.
- **Border Radius:** Rounded-xl (12pt) for cards; Rounded-2xl (16pt) for hero containers; Rounded-full for pill buttons.

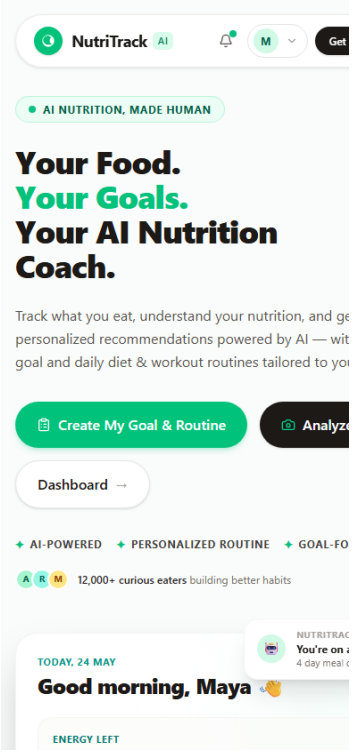
2. Landing & Authentication Flow (`/` & `/login`)

Initial entry point presenting feature highlights, quick-action cards, and hybrid login/registration.

Desktop Viewport (1280x950)



Mobile Viewport (390x844)



Key UI Components & Capabilities:

- **Hero Banner:** Dynamic greeting with one-tap navigation to Routine Generator and Food Scanner.
- **Interactive Preview Cards:** Visual teasers for Energy Gauge, Macro Bar, and Fasting Calendar.
- **Authentication Gateway:** Seamless switch between Guest mode and Cloud Account login.
- **Quick-Action Bar:** 1-tap shortcuts to Scan Meal, Generate Routine, and View Daily Diary.

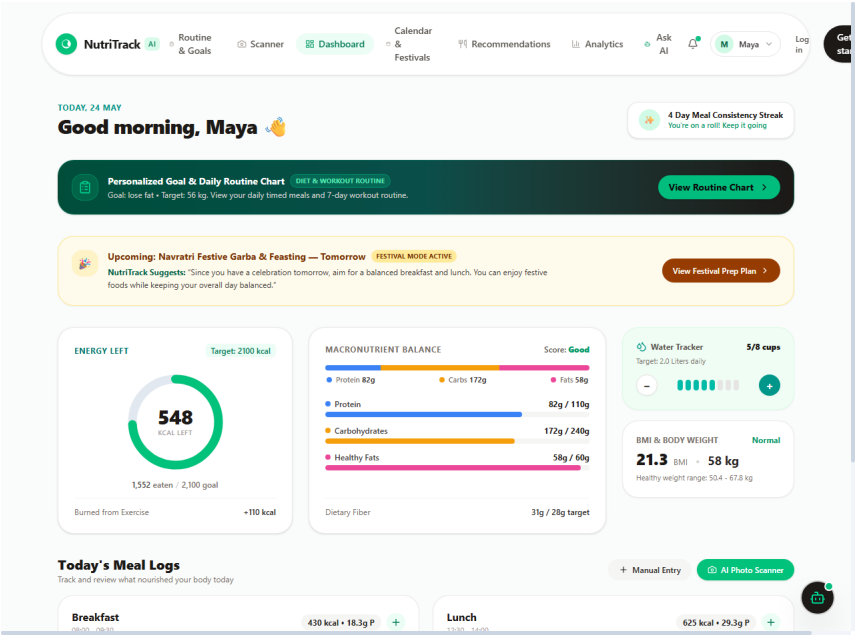
Technical Migration Directives:

- **Expo Routing:** Set `app/index.tsx` to automatically redirect authenticated users to `(tabs)/dashboard`.
- **Guest Mode:** Initialize a default `guest-user` profile in `AsyncStorage` when user skips login.
- **Component Structure:** Use React Native `` with `showsVerticalScrollIndicator={false}`.
- **Animations:** Implement subtle card scale-up on tap using `react-native-reanimated`.

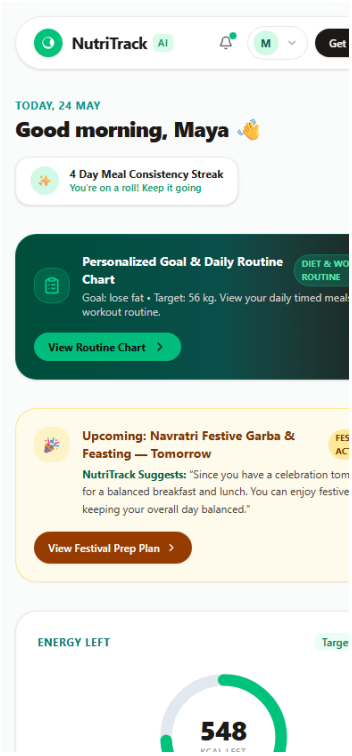
3. Nutrition Dashboard & Energy Gauge (`/dashboard`)

Central daily tracking screen showing calorie balance, macronutrient distribution, and logged meal history.

Desktop Viewport (1280x950)



Mobile Viewport (390x844)



Key UI Components & Capabilities:

- **Radial Energy Gauge:** Visualizes calories consumed vs target, with color transitions (Emerald, Amber, Rose).
- **Segmented Macro Split Bar:** Shows Protein, Carbohydrates, and Fats progress in grams and percentages.
- **Daily Meal Stream:** Categorized feed of Breakfast, Lunch, Dinner, and Snacks with photo thumbnails.
- **Interactive Water Tracker:** Quick-tap increment buttons (+250ml, +500ml) with glass visualizer.
- **Adherence Score Card:** Displays real-time 0-100 score (Exceptional, Good, Moderate, Needs Attention).

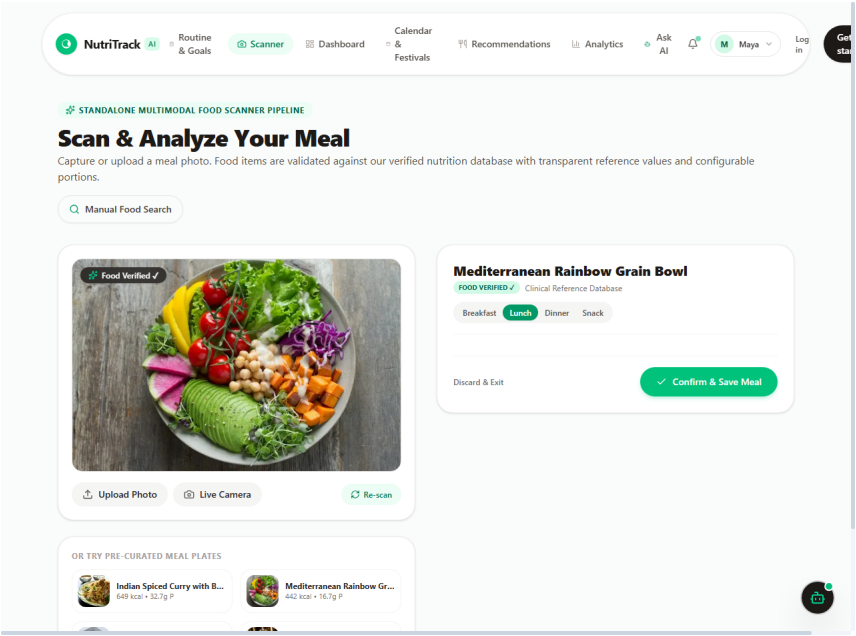
Technical Migration Directives:

- **Gauge Migration:** Use `react-native-svg` with animated `strokeDashoffset` interpolating current calories.
- **State Persistence:** Listen to `AsyncStorage` updates for `nutri_meals` to trigger instant reactive re-renders.
- **Pull-to-Refresh:** Wrap container in React Native `` to re-sync meals and profile from backend.
- **Haptic Feedback:** Trigger `Haptics.impactAsync(Haptics.ImpactFeedbackStyle.Light)` on water button taps.

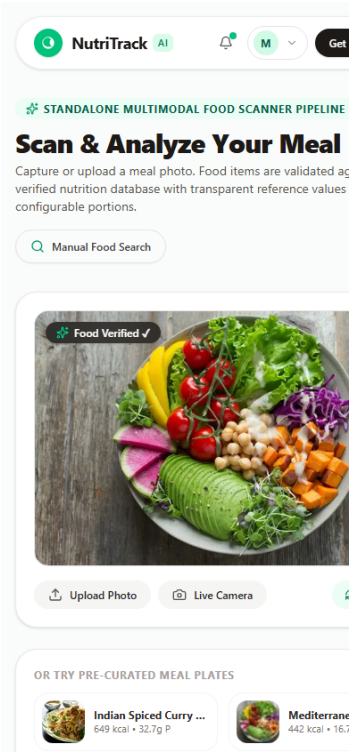
4. AI Food Vision & Camera Scanner (`/scanner`)

Instant meal identification and nutrient breakdown powered by OpenCLIP zero-shot inference.

Desktop Viewport (1280x950)



Mobile Viewport (390x844)



Key UI Components & Capabilities:

- **Dual Capture Modes:** Direct camera viewfinder capture or photo gallery file upload.
- **Real-Time Status Feedback:** Displays model status (accepted, needs_confirmation, rejected_non_food, low_quality).
- **Top Candidate Chips:** 1-tap selectable chips for ambiguous or competitive food classifications.
- **Portion & Serving Sizer:** Dropdown / chip selector for standard serving sizes (e.g. 1 medium, 1 bowl, 100g).
- **Nutrient Breakdown Table:** Displays Calories, Protein, Carbs, Fats, Fiber, Sodium, and Minerals.
- **Save to Diary Button:** 1-click commit into the user's active daily meal log.

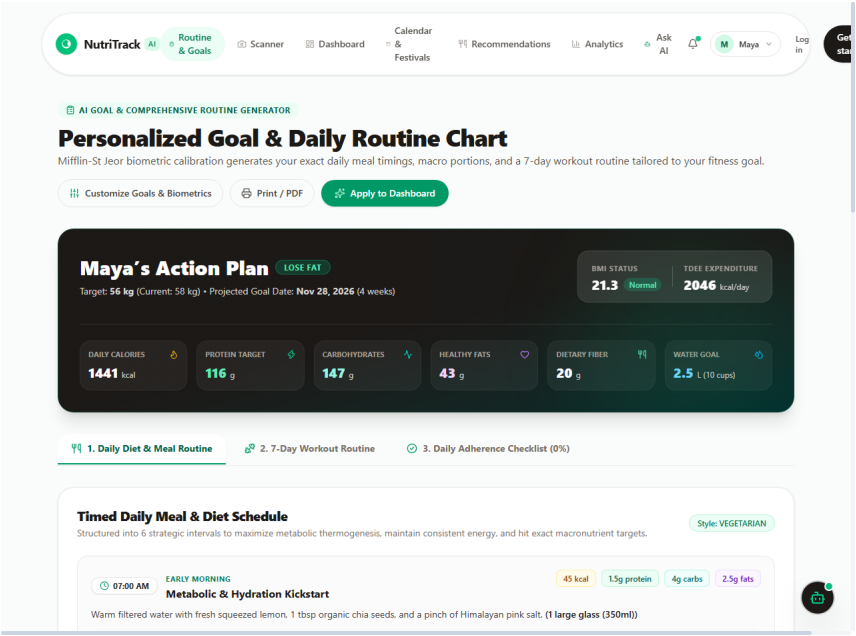
Technical Migration Directives:

- **Camera Integration:** Use `expo-camera` or `expo-image-picker` with `quality: 0.85` and `allowsEditing: true`.
- **Multipart Payload:** Format native image object as `{ uri, name: 'meal.jpg', type: 'image/jpeg' }` inside `FormData`.
- **API Endpoint:** Target `POST /api/scan-food` on the Next.js server proxy.
- **Serving Re-calculation:** Selecting a new portion size calls `POST /api/scan-food` with `{ foodId, grams }`.

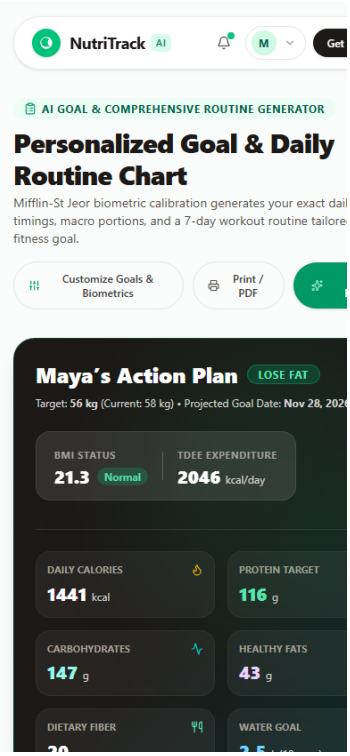
5. Goal & Routine Generator (`/routine`)

Clinically calculated fitness and diet planning engine producing customized 7-day workout and meal plans.

Desktop Viewport (1280x950)



Mobile Viewport (390x844)



Key UI Components & Capabilities:

- **Biometric Intake Wizard:** Age, gender, weight (kg), height (cm), target weight, and target pace selector.
- **Dietary & Workout Preferences:** Vegetarian, Vegan, Keto, Balanced; Gym, Dumbbells, or Bodyweight.
- **Clinical Biometrics Card:** Displays BMI, BMR (Mifflin-St Jeor), TDEE, Calorie Deficit/Surplus, and Target Date.
- **Timed 6-Meal Daily Plan:** Breakfast (08:00), Snack (11:00), Lunch (13:30), Pre-Workout (17:00), Dinner (20:30).
- **7-Day Exercise Schedule:** Detailed sets, reps, and exercise cards for Push, Pull, Legs, and Active Recovery.
- **Daily Habit Checklist:** Water target (ml), sleep requirement (8h), daily steps (10k), and protein pacing.

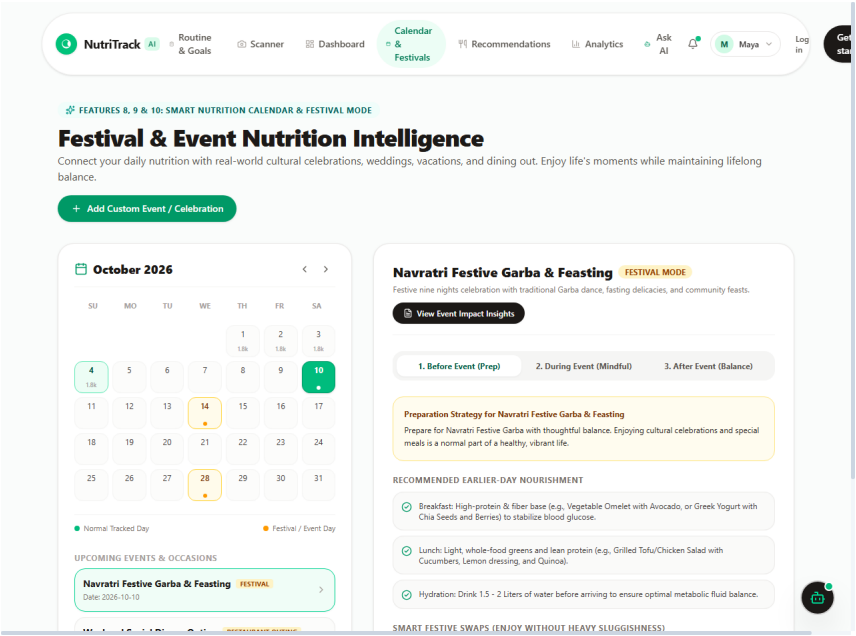
Technical Migration Directives:

- **Logic Reuse:** Directly copy `src/lib/routineEngine.ts` into the mobile project. Zero rewrites needed.
- **State Persistence:** Store generated plan in `AsyncStorage` under `nutri_routine`.
- **Interactive Toggles:** Allow users to check off daily habit checkboxes with persistent boolean flags.
- **Export / Share:** Use `expo-sharing` to allow users to export their routine plan as text or PDF.

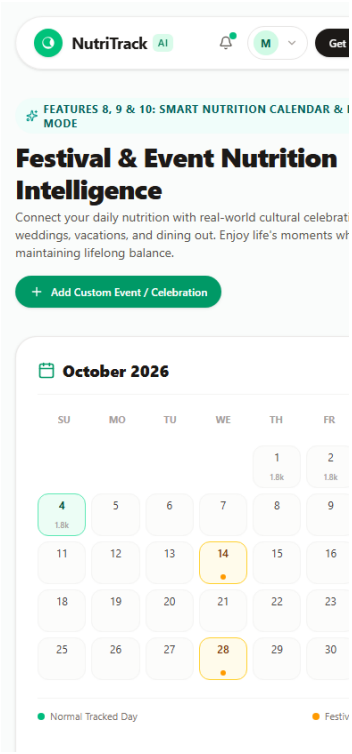
6. Cultural & Festival Calendar (`/calendar`)

Adaptive nutrition engine tailoring daily calorie and dietary recommendations to cultural festivals and fasting.

Desktop Viewport (1280x950)



Mobile Viewport (390x844)



Key UI Components & Capabilities:

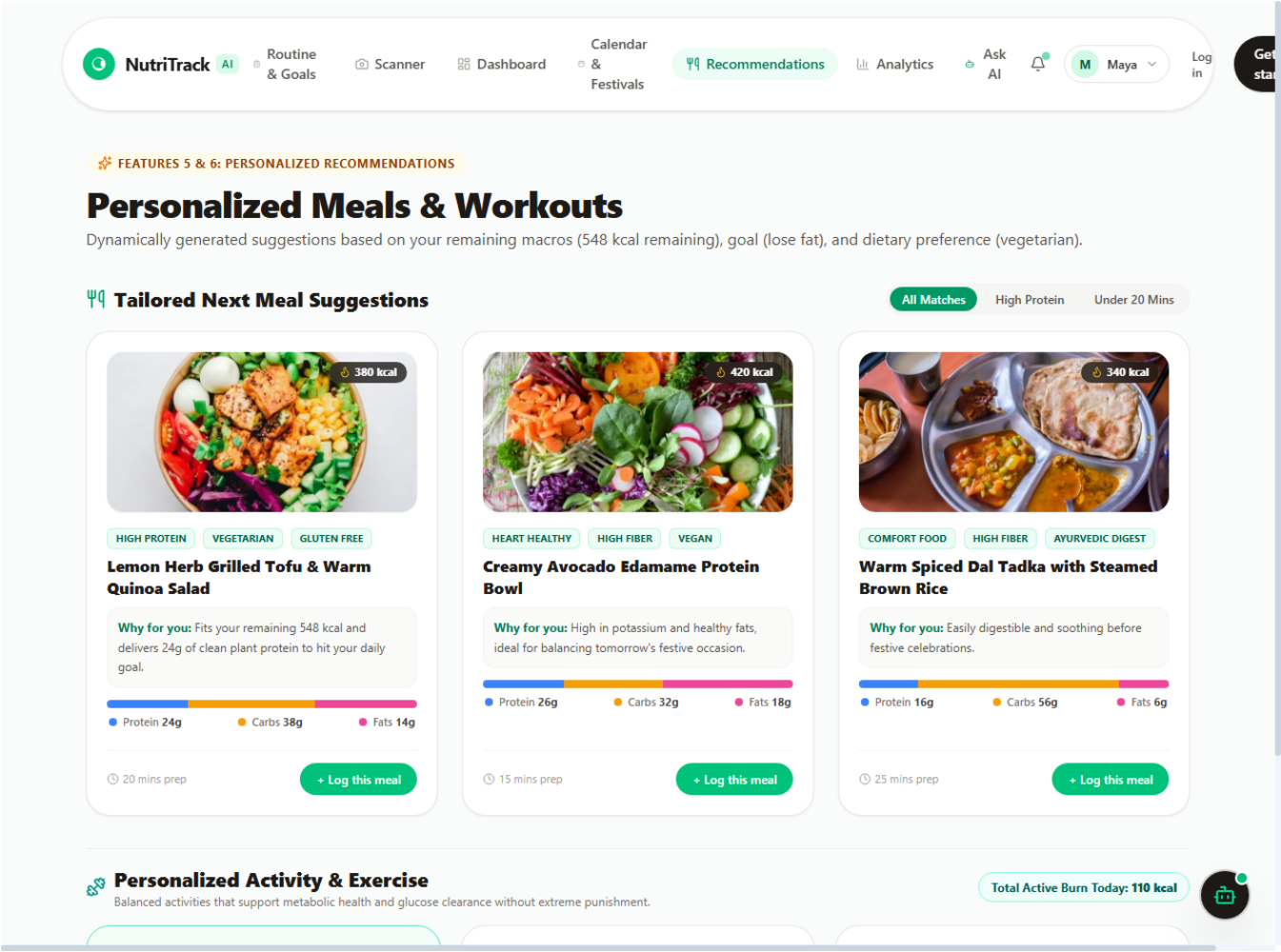
- **Monthly Interactive Calendar:** Date grid highlighting fasting days, religious observances, and feasts.
- **Festival Detail Drawer:** Shows event significance, permitted foods, and restricted items (e.g. Navratri grains).
- **Fasting Nutrition Advisor:** Specialized macro distribution for Ekadashi, Navratri, Ramadan, and Karwa Chauth.
- **Caloric Buffering System:** Automatically schedules minor calorie deficits prior to major celebrations.

Technical Migration Directives:

- **Logic Reuse:** Directly copy `src/lib/festivalEngine.ts` to power festival dates and fasting rules.
- **Calendar Component:** Implement using `react-native-calendars` or custom lightweight 7x5 flex grid.
- **Notification Alerts:** Use `expo-notifications` to schedule alerts on the morning of recognized fasting days.

7. Smart Recommendations & Guidance (`/recommendations`)

Algorithmic nutritional insights derived from the user's aggregated food logs and biometric deficiencies.



Key UI Components & Capabilities:

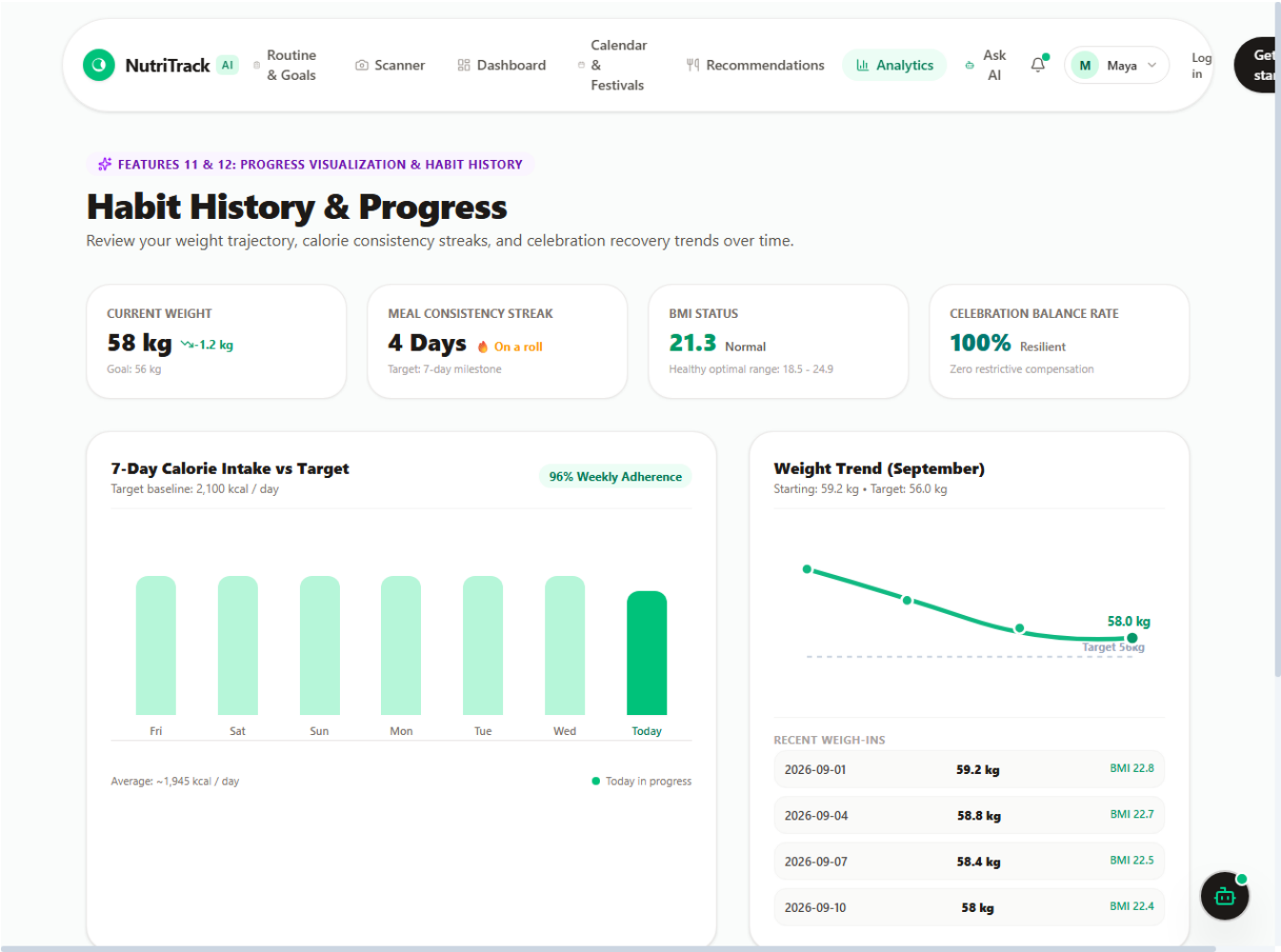
- **Deficiency Warning Banners:** Identifies low iron (<12mg), low fiber (<20g), or excessive sodium intake.
- **Actionable Food Swaps:** Suggests nutrient-dense alternatives (e.g. spinach + lemon for non-heme iron absorption).
- **Macronutrient Ratio Insights:** Compares actual protein-to-carbs ratio against target goal.
- **Hydration & Recovery Guidance:** Suggests electrolyte timing based on recorded workout frequency.

Technical Migration Directives:

- **Data Processing:** Evaluated dynamically by passing `nutri_meals` to `aggregateDailyNutrition()`.
- **Layout:** Card list rendered via React Native `` with pull-to-refresh.
- **Visual Indicators:** Color-coded badges (Emerald for optimal, Amber for warning, Sky for tips).

8. Analytics & Historical Trends (`/analytics`)

Comprehensive 7-day and 30-day nutritional charts tracking progress toward caloric and weight goals.



Key UI Components & Capabilities:

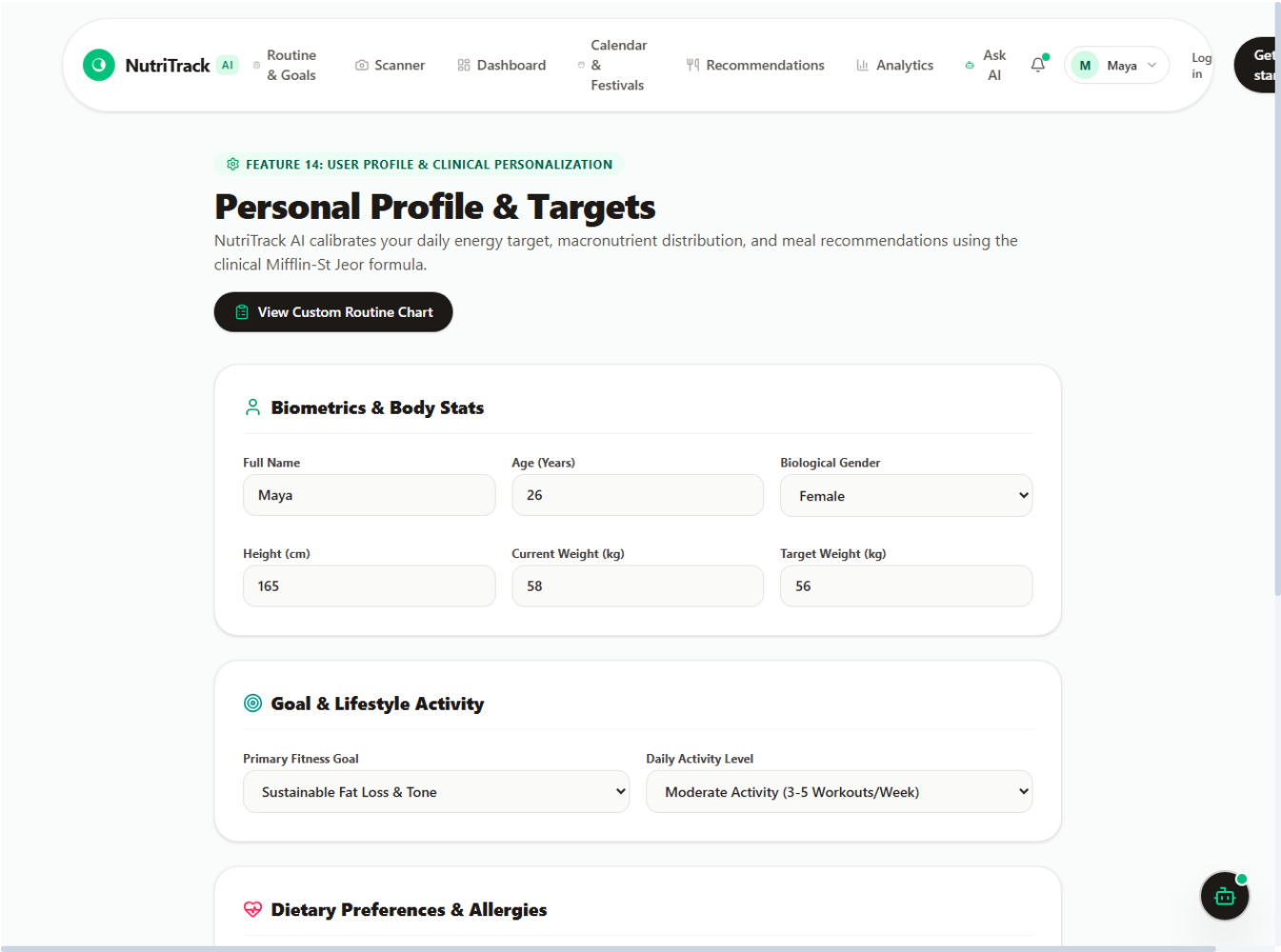
- **7-Day Calorie Adherence Chart:** Bar/line chart illustrating daily consumption against the target threshold.
- **Macronutrient Distribution Breakdown:** Donut chart showing cumulative protein, carbs, and fats proportions.
- **Weight Progress Curve:** Line graph plotting weigh-ins against the target milestone date.
- **Nutritional Quality Scorecard:** Weekly average score summary.

Technical Migration Directives:

- **Charting Library:** Use `react-native-gifted-charts` or `react-native-chart-kit` for performant native rendering.
- **Data Aggregation:** Group meals by date (`YYYY-MM-DD`) and sum calories and macros.
- **Theme Integration:** Supply NutriTrack theme colors (`#059669`, `#0284c7`, `#f59e0b`) to chart datasets.

9. Profile & Biometric Settings (`/profile`)

Personal health configuration, biometric updater, and account management hub.



Key UI Components & Capabilities:

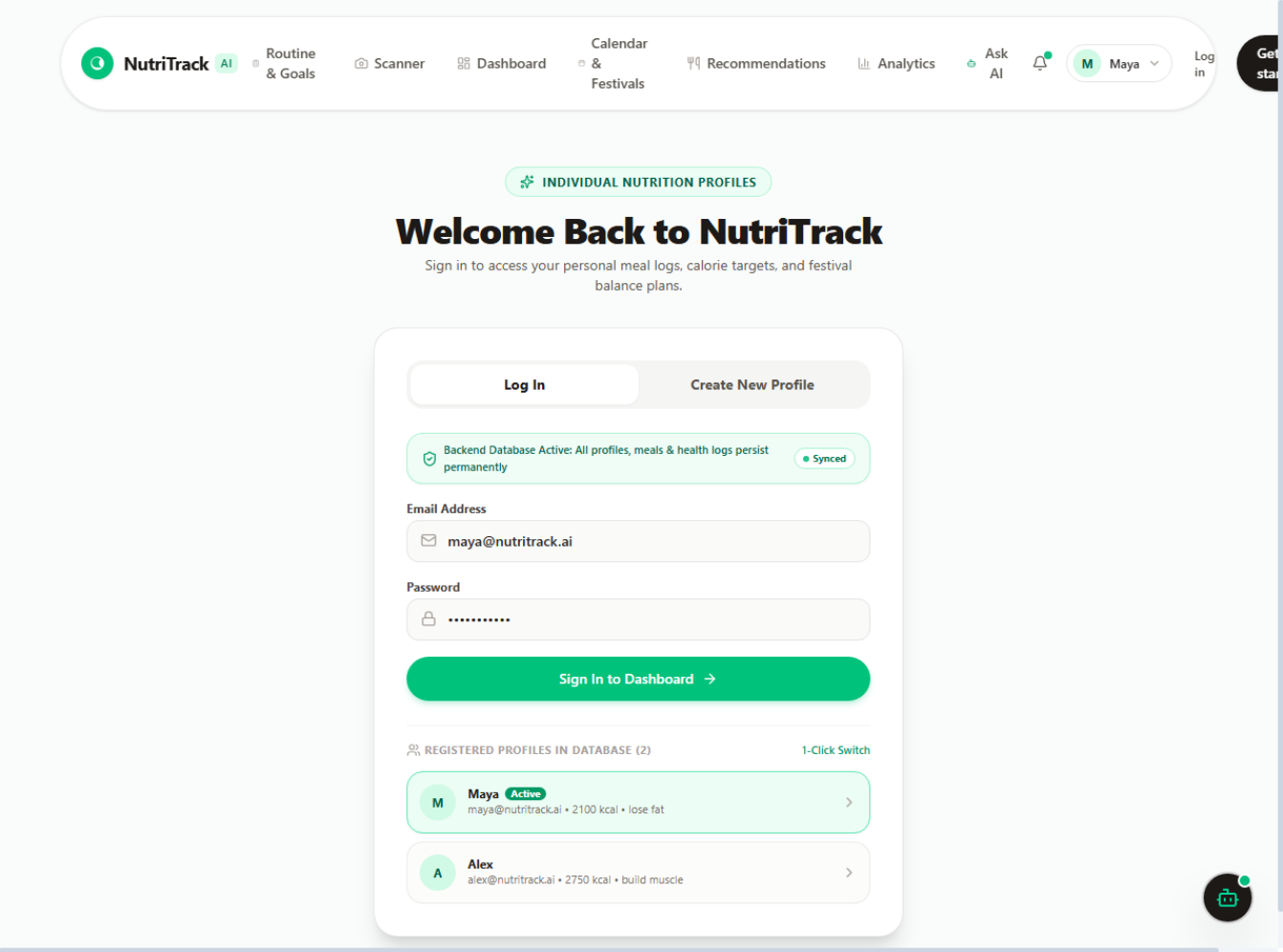
- **User Identity Header:** Avatar, name, account tier, and registered email address.
- **Biometric Editor:** Quick inputs for Weight (kg), Height (cm), Age, Gender, and Activity Level.
- **BMI Gauge & Ideal Weight Range:** Visual slider showing current BMI category and clinically healthy weight bounds.
- **Dietary & Allergy Tags:** Multi-select chips for Vegetarian, Vegan, Keto, Nut Allergy, Gluten Sensitivity.
- **Session Controls:** Clear local cache, sync to cloud, or log out of account.

Technical Migration Directives:

- **Form Management:** Use controlled state or `react-hook-form` for validated numerical inputs.
- **Real-time Math:** Changing weight or height immediately triggers `calculateBMI()` and updates the gauge.
- **Sync Mechanism:** Tapping 'Save Profile' writes to `AsyncStorage` and dispatches `POST /api/user/sync` if online.

10. Authentication & Security (`/login`)

User authentication interface with email/password login, account registration, and guest access.



Key UI Components & Capabilities:

- **Tabbed Interface:** Smooth toggle between Login and Sign Up views.
- **Quick Demo Access:** 1-tap pre-fill button for evaluating demo account (`demo@nutritrack.ai`).
- **Input Validation:** Validates email format and minimum 6-character password complexity.
- **Guest Access Link:** Allows users to bypass cloud authentication and use local storage exclusively.

Technical Migration Directives:

- **Keyboard Handling:** Wrap form in `KeyboardAvoidingView` with `behavior={Platform.OS === 'ios' ? 'padding' : 'height'}`.
- **Secure Text Entry:** Toggle password visibility via eye icon using local state.
- **API Proxy:** Dispatches `POST /api/auth/login` or `POST /api/auth/register`.

11. Mobile Interaction & Haptics Specification

To achieve a world-class native feel, Replit AI should integrate the following interaction standards:

User Action	Visual Feedback	Haptic Trigger	State Effect
Tap 'Scan Meal' CTA	Button scales down (0.97x)	`Haptics.impactAsync(Light)`	Opens camera viewfinder
Tap Water (+250ml)	Water level ripples up	`Haptics.impactAsync(Medium)`	Increments hydration count + saves
Confirm Scanned Food	Chip highlights in emerald	`Haptics.notificationAsync(Success)`	Loads portion & nutrition data
Save Meal to Diary	Success checkmark toast	`Haptics.notificationAsync(Success)`	Updates Energy Gauge immediately
Pull-to-Refresh	Emerald activity spinner	`Haptics.impactAsync(Light)`	Re-syncs profile & daily meals

Conclusion & Engineering Sign-Off

This specification represents the complete visual and behavioral contract for the NutriTrack mobile application. By following the provided component hierarchies, API schemas, and mathematical engines, Replit AI will achieve 100% parity with the existing web platform while delivering a polished, high-performance native iOS and Android experience.